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H_∞ gain-scheduled dynamic output feedback control with transient performance applied to electrical microgrid

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ABSTRACT

This paper introduces an approach for designing gain-scheduled dynamic output feedback controllers for continuous-time Linear Parameter-Varying (LPV) systems. The aim is to improve transient response by incorporating both D -stability and H_∞ criteria into the synthesis conditions. We achieve this by employing changes of variables and congruence transformations, which lead to new synthesis conditions expressed as linear matrix inequalities. Additional fine-tuned scalar parameters can be used to enhance further the controller performance in terms of less conservative H_∞ guaranteed costs. To assess the effectiveness of our proposed synthesis procedure, we conduct computational experiments in the context of a microinverter-based distributed power generation system. These experiments take into account real-world operational characteristics, providing a comprehensive evaluation of our approach. The results demonstrate that the designed controller effectively ensures the desired closed-loop system behavior, even when dealing with state noise and discretization errors originating from digital implementation.

1. Introduction

The theory of robust control design for power systems faces several challenges. It must not only comply with robust stability but also account for various additional constraints, including transient performance, time response, the stiff behavior of its states, parametric variation, and any other limitations imposed by the physical application [1–4]. Among the synthesis strategies for Linear Time-Invariant (LTI) systems dealing specifically with the closed-loop transient response, the formulations based on the concept of D -stability [5–8], which presents great flexibility to address pole location, stand out. Moreover, under certain assumptions, this technique can be adapted to deal with Linear Parameter-Varying (LPV) systems, allowing to design parameter-dependent controllers, also known as *Gain-Scheduled* (GS) controllers [9,10]. D -stability and related control techniques are suitable to address robust control using the Lyapunov stability theory, in general resulting in convex optimization problems expressed in terms of constraints in the form of Linear Matrix Inequalities (LMIs) [11], which can be efficiently solved by algorithms based on interior point methods [12,13].

In the robust control literature, LMI-based solutions address transient performance constraints. For example, in [14], a fuzzy static output-feedback (SOF) structure associated with D -stability is employed, resulting in a formulation based on LMI conditions

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using the classical framework from [15] to address the autonomous vehicle control problem. In [9], the design of SOF controllers is presented utilizing two different approaches: state derivative feedback and output-feedback, with a pole placement strategy similar to that in [16]. Furthermore, in both [17,18], a dynamic output-feedback (DOF) structure is utilized, and the D -stability constraints are also based on [15]. In [17], an optimal reset strategy is provided to improve a specific problem, whereas in [18], Cooperative Linear Output Regulation (CLOR) is proposed. It is worth noting that in the aforementioned papers, the authors focus on synthesis conditions performed in the continuous-time domain. Finally, [18] considers a direct input–output matrix to generate a regulation function. In this sense, by including exogenous input matrices, it is possible to enhance the generality of the system modeling.

When it comes to designing a controller for a class of LPV systems, there are two primary approaches to consider: the GS controller and the robust controller. The GS controller can accommodate parameter variations, generally leading to improved closed-loop performance. In contrast, the robust controller operates without any knowledge of the time-varying parameters, resulting in more conservative solutions. In the existing literature, various techniques have been proposed to mitigate the conservatism in solutions for LPV systems [19–25]. Moreover, different authors have worked on generating improved (less conservative) LMI-based techniques using different types of relaxations, such as the ones based on scalar searches [26–29].

Regarding applications, a microgrid system is a widely recognized and extensively utilized electrical power network, serving as the link between Distributed Generation (DG) units, the main electrical grid, and industrial or residential loads, all interconnected through a Point of Common Coupling (PCC). In the context of a typical microgrid power system, stringent requirements must be met. These requirements include the need for rapid response and robust damping characteristics while actively accounting for external disturbances. All of these factors aim to ensure the continuous stability of voltage and current conditions at the PCC. Furthermore, it is common for the system's controller to derive control signals from potentially noisy or imprecise state or output signals. In such applications, the overarching objective is to maintain system stability while concurrently optimizing transient performance. While performance criteria based on the H_∞ and H_2 norms have been extensively studied using various relaxation techniques [22,26,30], the consideration of transient performance has received less attention and deserves further investigation. This exploration aims to improve the closed-loop transient response while ensuring robust stability and effective disturbance rejection.

This paper introduces a novel LMI-based synthesis condition for designing DOF-GS controllers tailored specifically for continuous-time LPV systems. This technique extends the existing literature to encompass both D -stability and H_∞ criteria simultaneously. The proposed procedure is formulated as an algorithm that combines LMIs with the search for scalar parameters, which can be used to enhance further the controller performance in terms of less conservative H_∞ guaranteed costs. The performance of the proposed controller is evaluated through computational experiments within a simulated microinverter-based distributed power generation system, considering realistic operational characteristics. Under the specified conditions, the resulting GS-DOF controller ensures an upper bound for the H_∞ norm of the closed-loop system and achieves the desired transient performance, even when dealing with state signal noise. Furthermore, it is demonstrated that when implementing this controller on a digital microcontroller, the adverse effects of discretization errors can be mitigated by treating them as measurement errors in the state signal.

The paper is organized as follows: In Section 2, we formulate the problem and introduce the framework used throughout the article. Section 3 presents the derivation of the synthesis conditions for the DOF controller that will be employed. The electrical power system under study and its model are presented in Section 4. The control application is discussed in Section 5, while simulations are performed in Section 6 with two case studies to illustrate the performance of the proposed method in comparison to existing conditions. Finally, Section 7 presents the conclusions and outlines future directions.

Notation. Lowercase and capital letters indicate, respectively, vectors (or scalars) and matrices. The set of real numbers is denoted by $\mathbb{R}$, while natural numbers are given by $\mathbb{N}$. For real matrices and vectors, the symbol $(\cdot)'$ represents transposition, $\text{He}(M)$ is used to indicate the operation $M + M'$, and the symbol $(\star)$ stands for symmetric blocks in square matrices. The Newton notation is used to denote the derivatives, where $\dot{f}(t)$ represents the time-derivative of the function $f(t)$. The expression $M > 0$ ($M < 0$) indicates that M is positive (negative) definite. Parameter-dependent matrices (with affine dependence on the time-varying vector $\theta(t)$) are structured in the form

$$X(\theta(t)) = X_0 + \sum_{i=1}^V X_i \theta_i(t),$$

where X_i , $i = 0, \dots, V$ are given matrices, and $\theta(t)$ is a vector of bounded time-varying parameters belonging to the set

$$\Theta = \{(\theta_1, \dots, \theta_V) \in \mathbb{R}^V : \underline{\theta}_j \leq \theta_j \leq \overline{\theta}_j, j = 1, \dots, V\},$$

for all $t \geq 0$, where the bounds $\underline{\theta}_j$ and $\overline{\theta}_j$ are known values.

2. Problem statement and preliminaries

Consider the following LPV continuous-time system

$$\mathcal{G} : \begin{cases} \dot{x}(t) = A(\theta(t))x(t) + B(\theta(t))u(t) + J(\theta(t))w(t), \\ z(t) = C_z(\theta(t))x(t) + D_z(\theta(t))u(t) + E_z(\theta(t))w(t), \\ y(t) = C_y(\theta(t))x(t) + E_y(\theta(t))w(t), \end{cases} \quad (1)$$

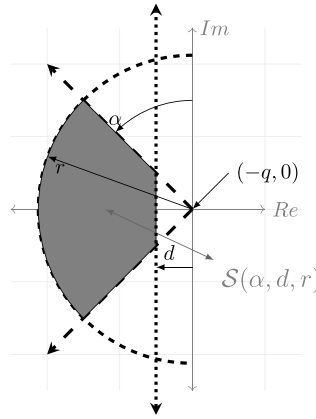


Fig. 1. Region desired $S(\alpha, r, d)$ on complex plane centered in $(-q, 0)$.

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^e$ is the control input, $w \in \mathbb{R}^p$ is the exogenous disturbance, $z \in \mathbb{R}^q$ is the regulated output, and $y \in \mathbb{R}^m$ is the measured output. In this paper, we consider a full-order DOF controller structured in the form

$$C : \begin{cases} \dot{x}_C(t) = A_C(\theta(t))x_C(t) + B_C(\theta(t))y(t), \\ u(t) = C_C(\theta(t))x_C(t) + D_C(\theta(t))y(t), \end{cases} \quad (2)$$

where $x_C \in \mathbb{R}^n$ are the states of the dynamic controller. Matrices $A_C(\theta(t))$, $B_C(\theta(t))$, $C_C(\theta(t))$, and $D_C(\theta(t))$ are parameter-dependent and have appropriate dimensions. The vector of time-varying parameters $\theta(t)$ is assumed to be measured (or estimated) in real-time. By defining the augmented state vector $\tilde{x}(t) = [x(t)' \ x_C(t)']'$, and connecting the DOF controller (2) to the system (1), we obtain the following closed-loop system

$$\mathcal{G}_C : \begin{cases} \dot{\tilde{x}}(t) = \tilde{A}(\theta(t))\tilde{x}(t) + \tilde{J}(\theta(t))w(t), \\ z(t) = \tilde{C}(\theta(t))\tilde{x}(t) + \tilde{E}(\theta(t))w(t), \end{cases} \quad (3)$$

where the closed-loop matrices are given by,

$$\begin{bmatrix} \tilde{A}(\theta(t)) & \tilde{J}(\theta(t)) \\ \tilde{C}(\theta(t)) & \tilde{E}(\theta(t)) \end{bmatrix} = \left[\begin{array}{cc|c} A(\theta(t)) + B(\theta(t))D_C(\theta(t))C_y(\theta(t)) & B(\theta(t))C_C(\theta(t)) & J(\theta(t)) + B(\theta(t))D_C(\theta(t))E_y(\theta(t)) \\ B_C(\theta(t))C_y(\theta(t)) & A_C(\theta(t)) & B_C(\theta(t))E_y(\theta(t)) \\ \hline C_z(\theta(t)) + D_z(\theta(t))D_C(\theta(t))C_y(\theta(t)) & D_z(\theta(t))C_C(\theta(t)) & E_z(\theta(t)) + D_z(\theta(t))D_C(\theta(t))E_y(\theta(t)) \end{array} \right].$$

The $\mathcal{H}_\infty$ guaranteed cost is a well-known control strategy to assure a bound γ to the energy gain from the disturbance input w to the regulated output z . The following lemma can be used to formulate the computation of an $\mathcal{H}_\infty$ guaranteed cost for system (3) using the concept of quadratic stability (the interested reader will find more details in [11]).

Lemma 1. *If there exists a symmetric positive definite matrix $P \in \mathbb{R}^{n \times n}$, and a scalar γ such that the following parameter-dependent inequality*

$$\begin{bmatrix} \text{He}(\tilde{A}(\theta)'P) & \star & \star \\ \tilde{J}(\theta)'P & -I & \star \\ \tilde{C}(\theta) & \tilde{E}(\theta) & -\gamma I \end{bmatrix} < 0 \quad (4)$$

holds for all $\theta \in \Theta$, then the closed-loop system (3) is quadratically stable and $\sqrt{\gamma}$ is an $\mathcal{H}_\infty$ guaranteed cost.

Note that if the controller matrices are known, the conditions of Lemma 1 are parameter-dependent LMIs, which can be solved efficiently through LMI relaxations [31,32].

In certain engineering problems, the primary focus is on the transient performance of the system. Therefore, it is essential to incorporate constraints associated with the controller design to ensure that the closed-loop system meets these requirements. In this context, the concept of D -stability plays a crucial role, allowing the placement of the closed-loop poles inside a convex region of the complex plane, as illustrated in Fig. 1. This region is defined by several parameters, including the inclination angle (α) of the boundary cone, the minimum desired decay rate (d), and the semicircle radius (r). Throughout this paper, we use the notation $S(\alpha, d, r)$ to specify this region. Although this technique was initially developed for LTI systems, extensions have been proposed in the literature to handle LPV systems. These extensions essentially assume that the time-varying parameters change slowly. The following Lemma [5,33] provides parameter-dependent inequalities that are sufficient to ensure that the eigenvalues of the closed-loop dynamic matrix $\tilde{A}(\theta)$, for fixed values of $\theta(t)$, remain inside the region S .

Lemma 2. Let $\alpha, r, d \in \mathbb{R}^+$ be given values. If there exists a symmetric positive definite matrix $P \in \mathbb{R}^{n \times n}$, such that the following parameter-dependent inequalities

$$\begin{aligned} \tilde{A}(\theta)P + PA(\theta)' + 2dP &< 0, \\ \begin{bmatrix} -rP & \star \\ qP + P\tilde{A}(\theta)' & -rP \end{bmatrix} &< 0, \\ (\Pi \otimes \tilde{A}(\theta))I_2P + I_2P(\Pi \otimes \tilde{A}(\theta))' &< 0, \end{aligned} \quad (5)$$

with

$$\Pi = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}, \quad I_2 = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix},$$

hold for all $\theta \in \Theta$, then the closed-loop system (3) is robustly stable and all the eigenvalues of the matrix $\tilde{A}(\theta)$, for frozen values of $\theta(t)$, belong to the $S(\alpha, r, d)$ region for all $\theta \in \Theta$.

The proof of Lemma 2 can be found in [15] and, if the controller matrices are given, then the inequalities in (5) are parameter-dependent LMIs.

This paper aims to present LMI-based synthesis conditions that ensure closed-loop performance in terms of $\mathcal{H}_\infty$ guaranteed costs and D -stability. To achieve this goal, we utilize the analysis conditions from Lemmas 1 and 2 to develop new synthesis conditions for the full-order DOF controller (2). The technical challenge lies in linearizing the nonconvex inequalities in (4) and (5) with minimal conservativeness. The next section presents the achieved results.

3. Main results

The following theorem presents the central contribution of this paper: a gain-scheduled $\mathcal{H}_\infty$ DOF control strategy with pole location in the region $S(\alpha, d, r)$ defined in terms of parameter-dependent LMIs, combined with linear searches for scalar parameters.

Theorem 1. Let $\alpha, r, d, \xi_i \in \mathbb{R}^+$, $i = 1, \dots, 3$, be given values with $\xi_1 > d$. If there exist symmetric positive definite matrices $\check{P}, \check{P}_1, \check{P}_2, \check{P}_3 \in \mathbb{R}^{2n \times 2n}$ partitioned as in (14), matrices R, G and $S \in \mathbb{R}^{n \times n}$, parameter-dependent matrices $F(\theta) \in \mathbb{R}^{n \times m}$, $K(\theta) \in \mathbb{R}^{e \times m}$, $Q(\theta) \in \mathbb{R}^n$ and $L(\theta) \in \mathbb{R}^n$, and a positive real scalar γ such that the following parameter-dependent LMIs

$$\begin{bmatrix} \xi \text{He}(\tilde{A}(\theta)) & \star & \star & \star \\ \check{P} - \xi \tilde{H}' + \tilde{A}(\theta) & -\text{He}(\tilde{H}) & \star & \star \\ \xi \check{B}(\theta)' & \check{B}(\theta)' & -\gamma I & \star \\ \check{C}(\theta) & 0 & \check{D}(\theta) & -I \end{bmatrix} < 0, \quad (6)$$

$$\begin{bmatrix} 2d\check{P}_1 + \xi_1 \text{He}(\tilde{A}(\theta)) & \star \\ \check{P}_1 - \xi_1 \tilde{H}' + \tilde{A}(\theta) & -\text{He}(\tilde{H}) \end{bmatrix} < 0, \quad (7)$$

$$\begin{bmatrix} -r\check{P}_2 & \star & \star \\ \check{P}_2 - \xi_2 \tilde{H}' & -\text{He}(\tilde{H}) & \star \\ \xi_2 \tilde{A}(\theta) & \tilde{A}(\theta) & -r\check{P}_2 \end{bmatrix} < 0, \quad (8)$$

$$\begin{bmatrix} \xi_3 \text{He}(\hat{A}(\theta)\Pi) & \star \\ \hat{P} - \xi_3 \hat{H}' + \hat{A}(\theta)\Pi & -\text{He}(\hat{H}) \end{bmatrix} < 0, \quad (9)$$

hold for all $\theta \in \Theta$, with

$$\begin{aligned} \hat{P} &= \begin{bmatrix} \check{P}_3 & 0 \\ 0 & \check{P}_3 \end{bmatrix}, \quad \hat{H} = \begin{bmatrix} \tilde{H} & 0 \\ 0 & \tilde{H} \end{bmatrix}, \\ \hat{A}(\theta) &= \begin{bmatrix} \tilde{A}(\theta) & 0 \\ 0 & \tilde{A}(\theta) \end{bmatrix}, \quad \Pi = \begin{bmatrix} \cos(\alpha)I_{2n} & \sin(\alpha)I_{2n} \\ -\sin(\alpha)I_{2n} & \cos(\alpha)I_{2n} \end{bmatrix}, \end{aligned}$$

and $\tilde{A}(\theta)$, $\tilde{B}(\theta)$, $\tilde{C}(\theta)$, $\tilde{D}(\theta)$, $\tilde{H}$, given by (15)–(19), respectively, then the closed-loop system (3) is robustly stabilized by the GS full-order DOF controller (2) with matrices

$$A_c(\theta) = Z^{-1} [Q(\theta)(S')^{-1} - RA(\theta)F(\theta)C_y(\theta) - RB(\theta)L(\theta)(S')^{-1} + RB(\theta)K(\theta)C_y(\theta)], \quad (10)$$

$$B_c(\theta) = Z^{-1} [F(\theta) - RB(\theta)K(\theta)], \quad (11)$$

$$C_c(\theta) = L(\theta)(S')^{-1} - K(\theta)C_y(\theta), \quad (12)$$

$$D_c(\theta) = K(\theta), \quad (13)$$

where $Z = (G - I)(S')^{-1} - R$. Additionally, $\sqrt{\gamma}$ is an $\mathcal{H}_\infty$ guaranteed cost for the closed-loop system, and the eigenvalues of $\tilde{A}(\theta)$, for frozen values of $\theta(t) \in \Theta$, belong to $S(r, d, \alpha)$.

Proof. Consider the following structured variables

$$H = \begin{bmatrix} R & Z \\ S^{-1} - R & -Z \end{bmatrix}, \quad \tilde{T} = \begin{bmatrix} I & S' \\ 0 & S' \end{bmatrix}, \quad \tilde{P} = \begin{bmatrix} P_1 & P_2 \\ P_2' & P_3 \end{bmatrix}.$$

By applying a congruence transformation on the Lyapunov matrix $\tilde{P}$, one gets

$$\tilde{T}' \tilde{P} \tilde{T} = \begin{bmatrix} W & J \\ J' & V \end{bmatrix} = \tilde{P}, \quad (14)$$

where

$$\begin{aligned} W &= P_1, \\ J &= P_1 S' + P_2 S', \\ V &= S(P_1 + P_2 + P_2' + P_3) S'. \end{aligned}$$

Also, by employing the change of variables

$$\begin{aligned} G &= RS' + ZS' + I, \\ K(\theta) &= D_C(\theta), \\ L(\theta) &= D_C(\theta)C_y(\theta)S' + C_C(\theta)S', \\ F(\theta) &= RB(\theta)D_C(\theta) + ZB_C(\theta), \\ Q(\theta) &= RA(\theta)S' + F(\theta)C_y(\theta)S' + RB(\theta)L(\theta) - RB(\theta)K(\theta)C_y(\theta)S' + ZA_C(\theta)S', \end{aligned}$$

one can rewrite the following matrices

$$\begin{aligned} \check{A}(\theta) &= \tilde{T}' \tilde{A}(\theta) H \tilde{T}, & \check{B}(\theta) &= \tilde{T}' H \tilde{B}(\theta), \\ \check{C}(\theta) &= \tilde{C}(\theta) \tilde{T}, & \check{D}(\theta) &= \tilde{D}(\theta), & \check{H} &= \tilde{T}' H \tilde{T}, \end{aligned}$$

as the following structured matrices:

$$\check{A}(\theta) = \begin{bmatrix} RA(\theta) + F(\theta)C_y(\theta) & Q(\theta) \\ A(\theta) + B(\theta)K(\theta)C_y(\theta) & A(\theta)S' + B(\theta)L(\theta) \end{bmatrix}, \quad (15)$$

$$\check{B}(\theta) = \begin{bmatrix} F(\theta)E_y(\theta) + RE(\theta) \\ B(\theta)K(\theta)E_y(\theta) + E(\theta) \end{bmatrix}, \quad (16)$$

$$\check{C}(\theta) = [C_z(\theta) + D_z(\theta)K(\theta)C_y(\theta) \quad C_z(\theta)S' + D_z(\theta)L(\theta)], \quad (17)$$

$$\check{D}(\theta) = E_z(\theta) + D_z(\theta)K(\theta)E_y(\theta), \quad (18)$$

$$\check{H} = \begin{bmatrix} R & G - I \\ I & S' \end{bmatrix}. \quad (19)$$

By multiplying (6) on the right by¹

$$\text{diag}(\tilde{T}^{-1}, \quad \tilde{T}^{-1}, \quad I, \quad I),$$

and on the left by its transpose, one obtains

$$Q + U' H' \mathcal{V} + \mathcal{V}' H U < 0, \quad (20)$$

with $U = [\bar{A} \quad -I \quad \bar{B} \quad 0]$, $\mathcal{V} = [\xi I \quad I \quad 0 \quad 0]$, and

$$Q = \begin{bmatrix} 0 & \star & \star & \star \\ \tilde{P} & 0 & \star & \star \\ 0 & 0 & -\gamma^2 I & \star \\ \tilde{C} & 0 & \tilde{D} & -I \end{bmatrix}.$$

Consider the following matrices as bases for the null spaces of U and $\mathcal{V}$ (i.e., $U' N_U = 0$ and $\mathcal{V}' N_V = 0$)

$$N_U = \begin{bmatrix} I & 0 & 0 \\ \bar{A} & \bar{B} & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}, \quad N_V = \begin{bmatrix} I & 0 & 0 \\ -\xi I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}.$$

¹ The dependence on the time-varying parameter θ is omitted in the proof to save space.

Pre and post multiplying (20) by $N'_{\mathcal{U}}$ and $N_{\mathcal{U}}$, one has

$$N'_{\mathcal{U}} Q N_{\mathcal{U}} = \begin{bmatrix} \text{He}(\tilde{A}' \tilde{P}) & \star & \star \\ \tilde{B}' \tilde{P} & -\gamma^2 I & \star \\ \tilde{C} & \tilde{D} & -I \end{bmatrix} < 0. \quad (21)$$

Note that (21) recovers (4) from a congruence transformation, that is, assures the validity of the *bounded real lemma* for the closed-loop LPV system using a quadratic Lyapunov function. Pre and post multiplying (20) by $N'_{\mathcal{V}}$ and $N_{\mathcal{V}}$, one has

$$N'_{\mathcal{V}} Q N_{\mathcal{V}} = \begin{bmatrix} -\xi \tilde{P} & \star & \star \\ 0 & -\gamma^2 I & \star \\ \tilde{C} & \tilde{D} & -I \end{bmatrix} < 0. \quad (22)$$

Since $\tilde{P} = \tilde{P}' > 0$, $\gamma > 0$ and $\xi > 0$, (22) is always feasible.

By multiplying (7) on the right by

$$\text{diag}(\tilde{T}^{-1}, \tilde{T}^{-1}),$$

and on the left by its transpose, one obtains

$$Q_1 + \mathcal{U}'_1 H' \mathcal{V}_1 + \mathcal{V}'_1 H \mathcal{U}_1 < 0, \quad (23)$$

with $\mathcal{U}_1 = [\tilde{A} \quad -I]$, $\mathcal{V}_1 = [\xi_1 I \quad I]$, and

$$Q_1 = \begin{bmatrix} -2d\tilde{P}_1 & \star \\ \tilde{P}_1 & 0 \end{bmatrix}.$$

Consider the following matrices as bases for the null spaces of $\mathcal{U}_1$ and $\mathcal{V}_1$, respectively,

$$N_{\mathcal{U}1} = \begin{bmatrix} I \\ \tilde{A} \end{bmatrix}, \quad N_{\mathcal{V}1} = \begin{bmatrix} I \\ -\xi_1 I \end{bmatrix}.$$

Pre and post multiplying (23) by $N'_{\mathcal{U}1}$ and $N_{\mathcal{U}1}$, yields

$$N'_{\mathcal{U}1} Q_1 N_{\mathcal{U}1} = \tilde{A}' \tilde{P}_1 + \tilde{P}_1 \tilde{A} + 2d\tilde{P}_1 < 0. \quad (24)$$

Note that (24) assures that the decay rate of the closed-loop system is bounded by d . Pre and post multiplying (23) by $N'_{\mathcal{V}1}$ and $N_{\mathcal{V}1}$, gives

$$N'_{\mathcal{V}1} Q_1 N_{\mathcal{V}1} = (2d - 2\xi_1) \tilde{P}_1 < 0. \quad (25)$$

Knowing that $\tilde{P}_1 = \tilde{P}'_1 > 0$, $d \geq 0$, and by hypothesis $\xi_1 > d$, (25) is always satisfied.

By multiplying (8) on the right by

$$\text{diag}(\tilde{T}^{-1}, \tilde{T}^{-1}, \tilde{T}^{-1}),$$

and on the left by its transpose, one obtains

$$Q_2 + \mathcal{U}'_2 H' \mathcal{V}_2 + \mathcal{V}'_2 H \mathcal{U}_2 < 0, \quad (26)$$

with $\mathcal{U}_2 = [0 \quad -I \quad \tilde{A}]$, $\mathcal{V}_2 = [\xi_2 I \quad I \quad 0]$, and

$$Q_2 = \begin{bmatrix} -r\tilde{P}_2 & \star & \star \\ \tilde{P}_2 & 0 & \star \\ 0 & 0 & -r\tilde{P}_2 \end{bmatrix}.$$

Consider the following bases for the null spaces of $\mathcal{U}_2$ and $\mathcal{V}_2$, respectively

$$N_{\mathcal{U}2} = \begin{bmatrix} I & 0 \\ 0 & \tilde{A} \\ 0 & I \end{bmatrix}, \quad N_{\mathcal{V}2} = \begin{bmatrix} I & 0 \\ -\xi_2 I & 0 \\ 0 & I \end{bmatrix}.$$

Pre and post multiplying (26) by $N'_{\mathcal{U}2}$ and $N_{\mathcal{U}2}$ yields

$$N'_{\mathcal{U}2} Q_2 N_{\mathcal{U}2} = \begin{bmatrix} -r\tilde{P}_2 & \star \\ \tilde{A}' \tilde{P}_2 & r\tilde{P}_2 \end{bmatrix} < 0. \quad (27)$$

Applying the Schur complement, the last inequality is equivalent to

$$-r^2 \tilde{P}_2 + \tilde{A}' \tilde{P}_2 \tilde{A} < 0,$$

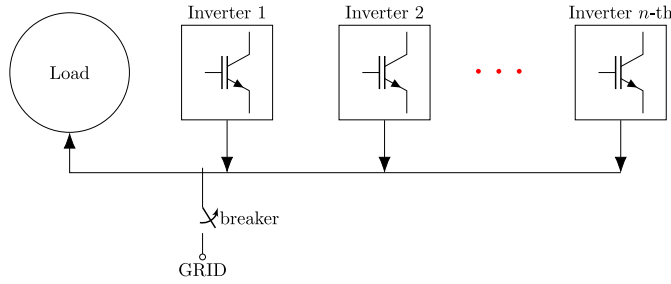


Fig. 2. Representation of a microgrid with n inverters and one load.

which assures, for frozen values of $\theta(t) \in \Theta$, that the eigenvalues of the closed-loop dynamic matrix $\tilde{A}(\theta)$ lie inside a disk of radius r . Pre and post multiplying (26) by $N'_{\mathcal{V}_2}$ and $N_{\mathcal{V}_2}$, yields

$$N'_{\mathcal{V}_2} Q_2 N_{\mathcal{V}_2} = \begin{bmatrix} -r\tilde{P}_2 - 2\xi_2\tilde{P}_2 & 0 \\ 0 & -r\tilde{P}_2 \end{bmatrix} < 0. \quad (28)$$

Knowing that $\tilde{P}_2 = \tilde{P}'_2 > 0$, $r > 0$, and $\xi_2 > 0$, (28) is always satisfied.

By multiplying (9) on the right by

$$\text{diag}(\hat{T}^{-1}, \hat{T}^{-1}), \quad \text{with } \hat{T} = \text{diag}(\tilde{T}, \tilde{T}),$$

and on the left by its transpose, one obtains

$$Q_3 + \mathcal{U}'_3 \hat{H}' \mathcal{V}_3 + \mathcal{V}'_3 \hat{H} \mathcal{U}_3 < 0, \quad (29)$$

with $\mathcal{U}_3 = [\mathcal{A}\Pi \quad -I]$, $\mathcal{V}_3 = [\xi_3 I \quad I]$, and

$$Q_3 = \begin{bmatrix} 0 & \mathcal{P} \\ \mathcal{P} & 0 \end{bmatrix},$$

where $\mathcal{A} = \text{diag}(\tilde{A}, \tilde{A})$ and $\mathcal{P} = \text{diag}(\tilde{P}_3, \tilde{P}_3)$. The following bases for the null spaces of $\mathcal{U}_3$ and $\mathcal{V}_3$ are considered

$$N_{\mathcal{U}_3} = \begin{bmatrix} I \\ \mathcal{A}\Pi \end{bmatrix}, \quad N_{\mathcal{V}_3} = \begin{bmatrix} I \\ -\xi_3 I \end{bmatrix}.$$

Pre and post multiplying (29) by $N'_{\mathcal{U}_3}$ and $N_{\mathcal{U}_3}$, gives

$$N'_{\mathcal{U}_3} Q_3 N_{\mathcal{U}_3} = \Pi' \mathcal{A}' \mathcal{P} + \mathcal{P} \mathcal{A} \Pi < 0, \quad (30)$$

Note that (30) can be rewritten as

$$\Psi \begin{bmatrix} \tilde{A}' & 0 \\ 0 & \tilde{A}' \end{bmatrix} \begin{bmatrix} \tilde{P}_3 & 0 \\ 0 & \tilde{P}_3 \end{bmatrix} + \begin{bmatrix} \tilde{P}_3 & 0 \\ 0 & \tilde{P}_3 \end{bmatrix} \begin{bmatrix} \tilde{A} & 0 \\ 0 & \tilde{A} \end{bmatrix} \Psi' < 0,$$

with

$$\Psi = \begin{bmatrix} \cos(\alpha)I_{2n_x} & -\sin(\alpha)I_{2n_x} \\ \sin(\alpha)I_{2n_x} & \cos(\alpha)I_{2n_x} \end{bmatrix}.$$

which, for frozen values of $\theta(t) \in \Theta$, assures that the eigenvalues of the closed-loop dynamic matrix $\tilde{A}(\theta)$ lie inside the cone delimited by angle α .

Pre and post multiplying (29) by $N'_{\mathcal{V}_3}$ and $N_{\mathcal{V}_3}$, gives

$$N'_{\mathcal{V}_3} Q_3 N_{\mathcal{V}_3} = -2\xi_3 \mathcal{P} < 0. \quad (31)$$

which is always satisfied because $\mathcal{P} > 0$ and $\xi_3 > 0$, finishing the proof. $\square$

It is important to emphasize that including the set of scalars $\xi, \xi_{1,2,3}$ introduces additional degrees of freedom in synthesizing the controller gains. Through optimization of these values, for example, via a linear search across a discrete set of values, it becomes possible to achieve enhanced (reduced) H_∞ guaranteed costs, albeit at the expense of increased computational demands associated with the linear search process. Regarding the practical implementation of the designed GS full-order DOF controller, note that real-time inversion of matrices is not required. This is because all inverse operations in Eqs. (10) to (13) are associated with parameter-independent matrices (an advantage of the proposed structured variables). Furthermore, it is worth noting that conditions (6) and (7) to (9) can be used independently. In other words, they can be employed to ensure only an H_∞ guaranteed cost or to perform pole placement within the S region.

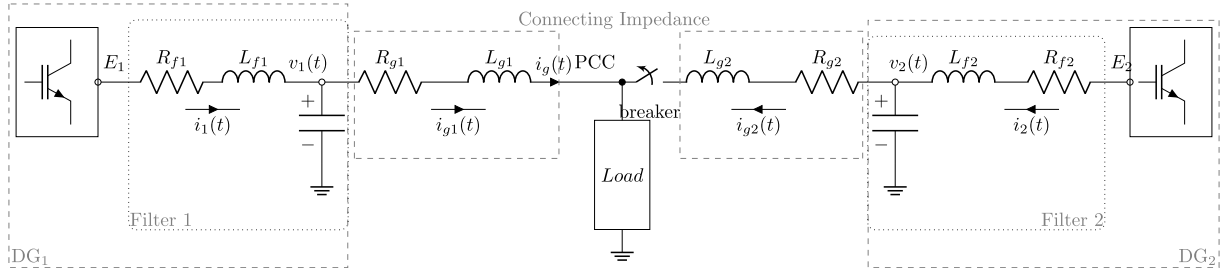


Fig. 3. Microgrid system composed of two DG units feeding to one unknown load. Note that DG_2 can be arbitrarily disconnected from the DG_1 -Load system by a breaker.

4. Electrical microgrid

A microgrid is an electrical network of power generators and loads. The microgrids can be grid-connected (connected to an available electrical grid) or isolated (as a local grid) as can be seen in Fig. 2. These systems are the focus of both academic and industrial investigations as they constitute a vital component of cutting-edge energy solutions [34]. This paper specifically explores an isolated microgrid featuring two inverters within a simulated test bench. The loads in the system are collectively represented as a single dynamic load. Fig. 3 shows two detailed Distributed Generation (DG) units connected to the load through the Point Common Couple (PCC). As power converters, three-phase voltage source inverters are used, and for a detailed understanding of their operation, one can refer to [35,36].

4.1. State-space model

The state-space model of the investigated microgrid is obtained through differential equations from the DG unit filter (see Fig. 3), where the abc -frame current and voltage of inverters are respectively modeled as $i_{j,abc} \in \mathbb{R}^{3 \times 1}$ and $E_{j,abc} \in \mathbb{R}^{3 \times 12}$

$$\begin{aligned} i_{j,abc}(t) &= i_{gj,abc}(t) + C_f \frac{dv_{j,abc}(t)}{dt}, \\ E_{j,abc}(t) &= L_f \frac{di_{j,abc}(t)}{dt} + R_f i_{j,abc}(t) + v_{j,abc}(t), \end{aligned} \quad (32)$$

where R_f , L_f and C_f are the resistor, inductance, and capacitance of the DG filter, $i_{gj,abc}(t) \in \mathbb{R}^{3 \times 1}$ and $v_{j,abc}(t) \in \mathbb{R}^{3 \times 1}$ are the current and voltage of the load. Note that $i_{gj,abc}(t)$ and $v_{j,abc}(t)$ are both vectors due to the three-phase nature of these quantities.

4.1.1. Model representation by static amplitude formulation

The Park and Clark transformations provide a method for modeling a three-phase system in the abc -frame as a system in the dq -frame. This transformation allows working in a rotational frame, holding the amplitudes constant. In contrast, in the abc -frame, the amplitudes are sinusoidal and vary with time, making it a static frame. The transformation process enables a static amplitude, allowing to obtain a state-space realization. For more detailed information about the transformation process and geometric analysis, please refer to the works in [37,38]. By applying the transformations, the differential equations below can represent a DG unit in the dq -frame.

$$dq\text{-frame} \begin{cases} \dot{v}_d(t) = \omega_0(t)v_q(t) + \frac{1}{C_f}i_d(t) - \frac{1}{C_f}I_{od}(t), \\ \dot{v}_q(t) = -\omega_0(t)v_d(t) + \frac{1}{C_f}i_q(t) - \frac{1}{C_f}I_{oq}(t), \\ \dot{i}_d(t) = \frac{R_f}{L_f}i_d(t) + \omega_0(t)i_q(t) + \frac{1}{L_f}m_d(t) - \frac{1}{L_f}v_d(t), \\ \dot{i}_q(t) = \frac{R_f}{L_f}i_q(t) - \omega_0(t)i_d(t) + \frac{1}{L_f}m_q(t) - \frac{1}{L_f}v_q(t), \end{cases} \quad (33)$$

where $\omega_0(t)$ represents the nominal operating frequency of the microgrid, $m_d(t)$ and $m_q(t)$ are the modulation indexes (switching signal function), $I_{oq}(t)$ and $I_{od}(t)$ represent the dq -coordinates of the current drawn by the load, the variables $v_d(t)$, $v_q(t)$, $i_d(t)$, and $i_q(t)$ are the dq -coordinate of the filter voltage and current, respectively. Consider $\tilde{x}(t) = [v_d(t) \ v_q(t) \ i_d(t) \ i_q(t)]'$ as the state vector, $\tilde{u}(t) = [m_d(t) \ m_q(t)]'$ as the input vector, $\tilde{w}(t) = [I_{od}(t) \ I_{oq}(t)]'$ as the disturbance input vector and $\tilde{z}(t) = [v_d(t) \ v_q(t)]'$ as the output. A state-space representation of Fig. 3 is given by

$$\begin{aligned} \dot{\tilde{x}}(t) &= A(t)\tilde{x}(t) + B\tilde{u}(t) + J\tilde{w}(t), \\ \tilde{z}(t) &= C_z\tilde{x}(t), \end{aligned} \quad (34)$$

² The subscript j refer to j th DG unit components.

where the matrices are

$$[A(t) \mid B \mid C'_z \mid J] = \left[\begin{array}{cccc|cccc|cc} 0 & \omega_0(t) & \frac{1}{C_f} & 0 & 0 & 0 & 0 & 1 & -\frac{1}{C_f} & 0 \\ -\omega_0(t) & 0 & 0 & \frac{1}{C_f} & 0 & 0 & 1 & 0 & 0 & -\frac{1}{C_f} \\ -\frac{1}{L_f} & 0 & \frac{R_f}{L_f} & \omega_0(t) & \frac{1}{L_f} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{L_f} & -\omega_0(t) & \frac{R_f}{L_f} & 0 & \frac{1}{L_f} & 0 & 0 & 0 & 0 \end{array} \right].$$

4.1.2. Model with time-varying parameters

Several control strategies, such as droop control, change the operating point of inverter systems. Droop control aims to compensate for the nominal frequency and reference voltage relative to the microgrid's active and reactive power demands. The nominal frequency is essential for synchronizing three-phase systems. An LPV model is adopted to account for these parameter variations.

Note that the nominal frequency is implicitly present in the matrix $A(t)$ provided in Eq. (34). When a control strategy compensating for reactive power is employed, the parameter $\omega_0(t)$ varies over time. To convert the model presented in (34) to an LPV affine model using (1), we choose a single time-varying parameter $\theta(t) = \omega_0(t)$. In this way, the new model becomes an LPV model in terms of the time-varying parameter $\theta(t)$.

4.2. Augmented control model

The control objective is to hold the voltage reference while one/two DG units supply load-current demand. To achieve this aim, it is necessary to include a reference tracking method. For this purpose, the following system is investigated

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} A(\theta(t)) & 0_{\tilde{n} \times \tilde{q}} \\ -C_z & 0_{\tilde{q} \times \tilde{q}} \end{bmatrix} x(t) + \begin{bmatrix} B \\ 0_{\tilde{q} \times \tilde{e}} \end{bmatrix} u(t) + \begin{bmatrix} J \\ 0_{\tilde{q} \times \tilde{p}} \end{bmatrix} w(t) + \begin{bmatrix} 0_{\tilde{n} \times \tilde{q}} \\ C_z \end{bmatrix} V_{ref}, \\ z(t) &= \begin{bmatrix} C_z \\ 0_{\tilde{q} \times \tilde{q}} \end{bmatrix} x(t), \end{aligned} \quad (35)$$

where $A(\theta(t)) = (A_0 + A_1\theta(t))$ with

$$A_0 = \begin{bmatrix} 0 & 0 & \frac{1}{C_f} & 0 \\ 0 & 0 & 0 & \frac{1}{C_f} \\ -\frac{1}{L_f} & 0 & \frac{R_f}{L_f} & 0 \\ 0 & -\frac{1}{L_f} & 0 & \frac{R}{L_f} \end{bmatrix}, \quad A_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}.$$

The state vector is $x(t) = [\tilde{x}(t)' \quad x_e(t)']'$, being $\tilde{x}(t)$ the states from (34), the input control vector is $u(t) = [\tilde{u}(t)' \quad 0]'$, being $\tilde{u}(t)$ the input control from (34), and the exogenous input is $w(t) = [\tilde{w}(t)' \quad 0]'$ being $\tilde{w}(t)$ the exogenous input from (34). The new state $x_e(t)$ represents the error $C_z V_{ref} - C_z \tilde{x}(t)$.

4.2.1. Modeling a piecewise continuous signal as disturbance

When a controller is implemented on a digital microcontroller, a digitalization error arises due to the sampling of the continuous signal. As a result, the signal behaves as a piecewise continuous signal, which introduces errors into the system. To address this issue, we consider the continuous-time measure of the states as the sampled discrete states, with the addition of a discretization error, resulting in:

$$x(t) = x(kT) + \zeta(t), \quad (36)$$

where $x(kT)$ represents the vector of sampled states, and $\zeta(t)$ is the error vector associated with the sampling period. If the standard deviation vector $\sigma \in \mathbb{R}^{n_x \times 1}$ of $\zeta(t)$ with respect to T is linear and known, then, the error $\zeta(t)$ can be rewritten as

$$\zeta(t) = \sigma \delta(t - kT),$$

where $\delta(t - kT)$ is the period of time between samples over which the error is accumulated. Taking the measured output $y(t)$ from (1) and applying the new definition of the states (36), a new measured output is obtained:

$$\begin{aligned} y(t) &= C_y (x(kT) + \sigma \delta(t - kT)) + E_y w(t), \\ &= C_y x(kT) + C_y \sigma \delta(t - kT) + E_y w(t). \end{aligned} \quad (37)$$

By considering $\tilde{w}(kT) = [w(t)' \quad \delta(t - kT)']' \in \mathbb{R}^{(n_w + n_x)}$, one has

$$y(t) = C_y x(kT) + [E_y \mid C_y \sigma] \tilde{w}(kT). \quad (38)$$

In this paper, we assume that the variation standard vector is $\sigma = 0.005 I_{n_x}$. The disturbance vector $w(t)$ does not affect the measure, i.e., $E_y = 0_{n_w}$. By using these definitions, we can operate the system in discrete-time with a controller designed in continuous-time while accounting for measurement errors.

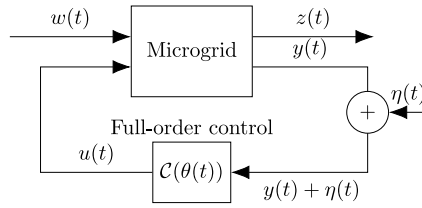


Fig. 4. General input-output scheme of proposed control topology.

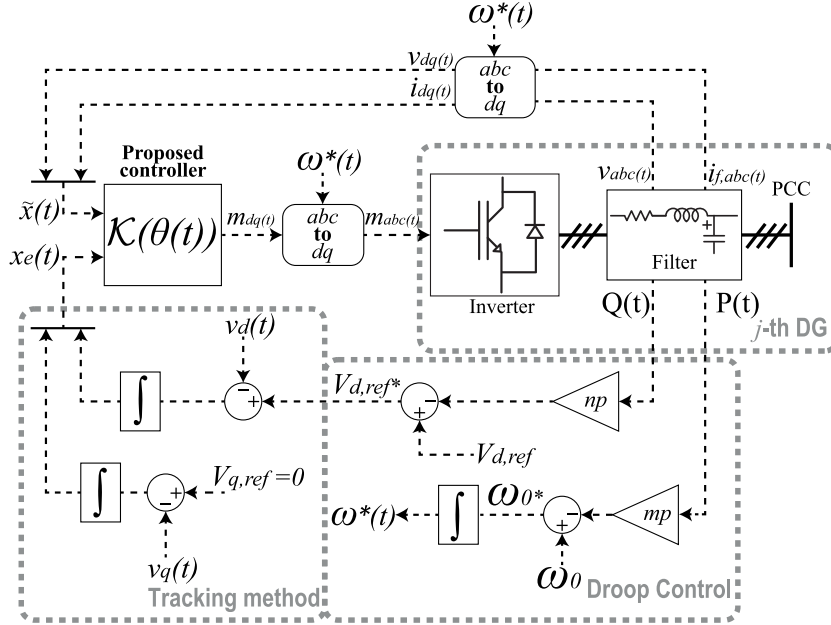


Fig. 5. Parts of the proposed control scheme: second level control (Droop area), tracking reference method (tracking area), and the proposed control. This scheme shows only one DG unit, but it is extensible to any other DG unit.

5. Control design for microgrid

In applications of this nature, measurement noise poses a challenge for classical control, which can be mitigated by investing in less noise-sensitive but more expensive instrumentation [39,40]. However, the proposed dynamic output feedback control strategy can operate with noisy signals while simultaneously filtering them to generate a control signal with the desired performance. The control strategy consists of two levels of control. The second level is a droop control that provides the voltage reference to the primary control and compensates for the nominal frequency. The primary control is the proposed approach, where the GS full-order DOF controller is obtained via the LMI conditions of Theorem 1. The general structure is depicted in Fig. 4.

This control scheme holds the output voltage over the reference voltage (tracking set-point) while rejecting the disturbance in the presence of a noisy state signal $(x(t) + \eta(t))$. Fig. 5 shows the detailed control structure, where $\omega^*(t)$ corresponds to synchronization angle for the dq -transformation, and the blocks abc -to- dq and dq -to- abc are the coordinate transformations. Particularly, the block abc -to- dq is used to convert (32) to (33), and the block dq -to- abc turns the control signal into abc -frame. The Droop Control zone contains a droop-based strategy widely used in electrical systems, whose proposition is adjusting the synchronization angle and the voltage reference, which is used for the primary control. The tracking method zone generates the $x_e(t)$ states based on the accumulation of the error.

Proposed design parameters: Each application has different design parameters for the region $S(\alpha, d, r)$. For electrical microgrids, some papers [4,33,41] propose using the following guidelines: the radius r must start at the origin ($q = 0$), and its length must be one-tenth of the maximum dynamics admissible stable on the real-axis of the s -plane, i.e., it must be at least ten times slower than the sampling rate (f_{sw}). A major r does not guarantee that the commutation dynamics are negligible. The choice of d depends on the minimum useful dynamics for the system. The electrical system operates with a ω_0 frequency (50 Hz or 60 Hz depending on the country), and the authors assume that five times ω_0 in rad s^{-1} is a reasonable minimum dynamics. The parameter α in Fig. 1 corresponds to the angle between the imaginary axis and the bounded imaginary amplitude with respect to the real axis. This constraint also can be seen as a concave cone to the left, where it is the limit of the imaginary dynamics. In the literature, this parameter is known as the

Table 1
Numerical values used in the numerical experiments.

(a) Parameters of the model.			(b) Design constraints for the LMIs in Eqs. (7) to (9).		
	Value	Unit		Value	Unit
R_f	0.1	Ω	f_{sw}	40	kHz
L_f	1.35	mH	ω_0	50	Hz
C_f	50	μF	d	$5(2\omega_0\pi)$	s^{-1}
R_g	0.2	Ω	r	$2f_{sw}\pi/10$	s^{-1}
L_g	0.3	mH	α	$\pi/4$	rad
f	[47, 53]	Hz	ξ	$[1, 2, \dots, 9] \otimes [10^0, 10^1, \dots, 10^6]$	
ω	$2f\pi$	rad s^{-1}	ξ_1	$[1, 2, \dots, 9] \otimes [10^3, 10^4, \dots, 10^6]$	
P_{max}	30	kW	ξ_2	$[1, 2, \dots, 9] \otimes [10^0, 10^1, \dots, 10^6]$	
Q_{max}	1	kVAr	ξ_3	$[1, 2, \dots, 9] \otimes [10^0, 10^1, \dots, 10^6]$	

damping factor and represents how oscillatory a response can be. The greater the angle, the smaller the oscillation. The literature suggests an angle over 45° for this reference frame. Table 1a presents the parameter values from the investigated system. Based on the dynamic constraint for $S(\alpha, d, r)$ region, Table 1b was created. The values presented in Table 1a are used when solving the LMI conditions of Theorem 1.

5.1. Synthesized controller

To synthesize a controller for the case study, the values from Table 1a and b are employed when solving the conditions of Theorem 1, yielding the controller $C(\theta(t))$ structured as in (39). This controller assures an $\mathcal{H}_\infty$ guaranteed cost given by γ . The parameters were found by performing a linear search in a discrete set of values logarithmically spaced, as informed in Table 1b, and the values that provided the best performance index were: $\xi = [10^4]$, $\xi_1 = [10^4]$, $\xi_2 = [10^5]$ and $\xi_3 = [10^4]$, yielding $\gamma = 14.67$. Note that $C(\theta(t))$ is a GS full-order DOF controller depending affinely on the time-varying parameter $\theta(t)$:

$$C(\theta(t)) = \begin{cases} \dot{x}_C(t) = (A_{C0} + A_{C1}\theta(t))x_C(t) + (B_{C0} + B_{C1}\theta(t))y(t), \\ u(t) = (C_{C0} + C_{C1}\theta(t))x_C(t) + (D_{C0} + D_{C1}\theta(t))y(t). \end{cases} \quad (39)$$

where matrices A_{C0} , A_{C1} , ..., D_{C1} are given by

$$A_{C0} = \begin{bmatrix} -6430.0 & -0.172 & 20.7 & 3.79 & 1.13 & -0.0286 \\ 0.171 & -6430.0 & -3.79 & 20.7 & 0.0183 & 1.26 \\ -1470.0 & -77.7 & -1.16 \times 10^4 & -103.0 & 3.35 \times 10^6 & -3600.0 \\ 75.8 & -1660.0 & 106.0 & -1.18 \times 10^4 & 1.26 \times 10^4 & 3.67 \times 10^6 \\ 0.784 & 0.0426 & 2.27 & 0.064 & -8640.0 & -8.78 \\ -0.0463 & 0.86 & -0.0772 & 2.26 & 13.0 & -8790.0 \end{bmatrix}$$

$$A_{C1} = \begin{bmatrix} -0.00536 & 2.49 \times 10^{-5} & 0.0176 & -5.57 \times 10^{-6} & -4.09 \times 10^{-4} & 2.22 \times 10^{-5} \\ -2.08 \times 10^{-5} & -0.00536 & 2.15 \times 10^{-6} & 0.0176 & -8.05 \times 10^{-6} & -6.89 \times 10^{-4} \\ -0.246 & -7.57 \times 10^{-4} & -0.795 & -0.0775 & -1840.0 & -13.8 \\ -0.00256 & 0.37 & 0.07 & -0.222 & 1.51 & -2910.0 \\ 7.35 \times 10^{-4} & 1.24 \times 10^{-5} & 0.00279 & 7.64 \times 10^{-5} & 0.234 & 0.0212 \\ -1.97 \times 10^{-5} & 6.65 \times 10^{-4} & -7.56 \times 10^{-5} & 0.00323 & -0.0207 & 0.497 \end{bmatrix}$$

$$B_{C0} = \begin{bmatrix} 6430.0 & 299.0 & 2.0 \times 10^4 & -3.79 & -1.53 & 0.0435 \\ -299.0 & 6430.0 & 3.79 & 2.0 \times 10^4 & -0.0359 & -1.57 \\ -1920.0 & -144.0 & -3460.0 & -139.0 & 8.98 \times 10^5 & -3830.0 \\ 143.0 & -2010.0 & 137.0 & -3430.0 & 3890.0 & 9.45 \times 10^5 \\ -1.85 & -0.0473 & -3.19 & -0.056 & 9030.0 & -2.61 \\ 0.0561 & -1.86 & 0.0733 & -3.4 & -4.55 & 9120.0 \end{bmatrix}$$

$$B_{C1} = \begin{bmatrix} 0.00536 & -2.49 \times 10^{-5} & -0.0176 & 5.57 \times 10^{-6} & 8.15 \times 10^{-4} & -1.91 \times 10^{-5} \\ 2.08 \times 10^{-5} & 0.00536 & -2.16 \times 10^{-6} & -0.0176 & 1.77 \times 10^{-5} & 8.14 \times 10^{-4} \\ 0.375 & 0.0197 & 0.98 & 0.0425 & 1810.0 & -2.58 \\ -0.019 & 0.618 & -0.0418 & 1.0 & 6.02 & 1730.0 \\ -3.3 \times 10^{-4} & -1.17 \times 10^{-5} & -0.00156 & -7.99 \times 10^{-5} & -1.17 & -3.08 \times 10^{-4} \\ 1.16 \times 10^{-5} & -4.59 \times 10^{-4} & 8.37 \times 10^{-5} & -0.0016 & 0.00234 & -1.17 \end{bmatrix}$$

$$C_{C0} = \begin{bmatrix} -1.98 & -0.105 & -7.01 & -0.142 & 4530.0 & -4.86 \\ 0.102 & -2.24 & 0.145 & -7.29 & 17.0 & 4950.0 \end{bmatrix}$$

$$C_{C1} = \begin{bmatrix} -3.25 \times 10^{-4} & -1.03 \times 10^{-6} & -0.00107 & -1.05 \times 10^{-4} & -2.49 & -0.0186 \\ -3.45 \times 10^{-6} & 5.06 \times 10^{-4} & 9.47 \times 10^{-5} & -2.96 \times 10^{-4} & 0.00203 & -3.93 \end{bmatrix}$$

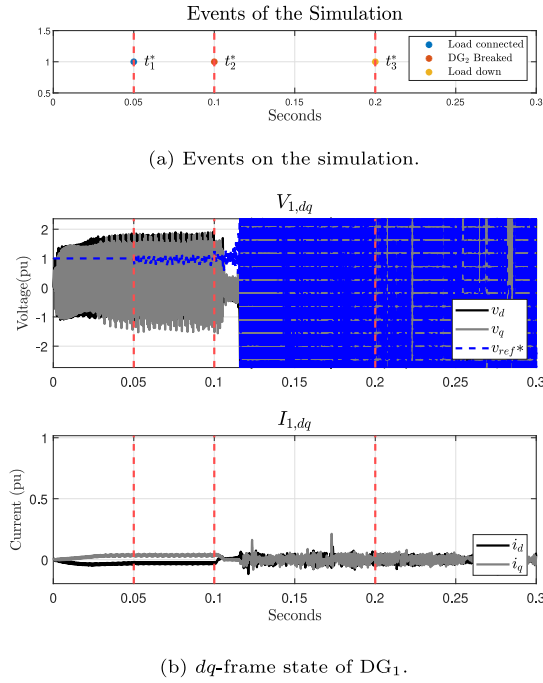


Fig. 6. Response considering only the $\mathcal{H}_\infty$ performance: (a) time marks associated with each event affecting the microgrid over time; (b) states on dq -frame of the DG_1 unit (dq -coordinate of the filter voltage and current, respectively). The controller used for this simulation can be seen in [Appendix A](#).

$$D_{C0} = \begin{bmatrix} -1.59 & -0.195 & -13.2 & -0.589 & 1210.0 & -5.18 \\ 0.193 & -1.72 & 0.586 & -13.2 & 5.26 & 1280.0 \end{bmatrix}$$

$$D_{C1} = \begin{bmatrix} 4.99 \times 10^{-4} & 2.66 \times 10^{-5} & 0.00132 & 5.77 \times 10^{-5} & 2.44 & -0.00348 \\ -2.57 \times 10^{-5} & 8.27 \times 10^{-4} & -5.67 \times 10^{-5} & 0.00135 & 0.00813 & 2.34 \end{bmatrix}$$

Line droop compensation control: This control is obtained from [\[42\]](#)

$$\begin{aligned} V_{d,ref}^* &= V_{d,ref} - n_p Q, & n_p &= \frac{V_{d,ref,max} - V_{d,ref,min}}{Q_{max}}, \\ \omega_0^* &= \omega_0 - m_p P, & m_p &= \frac{\omega_{0,max} - \omega_{0,min}}{P_{max}}. \end{aligned} \quad (40)$$

where m_p and n_p are known as the droop coefficients. By employing the values presented on [Table 1a](#) and considering an acceptable variation of 2% in $V_{d,ref}$ and 1% in ω_0 , $m_p = 0.0003$ and $n_p = 0.0062$ were obtained.

6. Case studies

Two experiments are presented in this section in order to illustrate the time response in two scenarios: using only $\mathcal{H}_\infty$ performance (only (6) is solved in [Theorem 1](#)) and using both $\mathcal{H}_\infty$ performance and D -stability (all conditions of [Theorem 1](#) are considered). The latter experiment evaluates the importance of considering the D -stability criterion. The first experiment shows the conduct of distinct operating modes, such as single DG unit operation, double DG units operating, and increasing load demand for a given controller. The second experiment is a direct comparison of the performance on frequency through a sweep in the sampling time of a zero-order holder, the period of the measurement error, and the noise frequency on the states.

6.1. Case study 1: Performance with/without D -stability on constant power load

For this time simulation, temporal events (stages) are considered:

- (i) at the beginning ($0 \leq t \leq t_1^*$), the system (both DG_1 and DG_2 units) feeds to a 10 kW resistive load;
- (ii) at $t_1^* = 0.05$ s a 10 kW-2 kVar dynamic Constant Power Load (CPL) load is connected;
- (iii) at $t_2^* = 0.1$ s, the DG_2 unit is disconnected;
- (iv) at $t_3^* = 0.2$ s, the dynamic load decreases to 50% of its initial value.

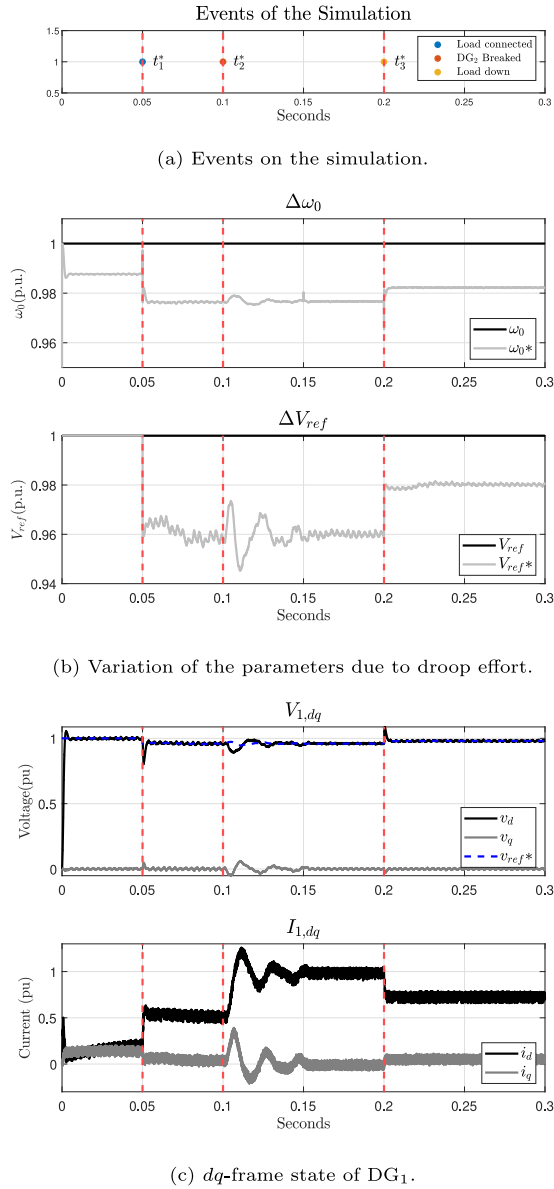


Fig. 7. Response considering both $\mathcal{H}_\infty$ and D -stability performance: (a) time marks associated with each event affecting the microgrid over time; (b) Variation of the parameters due to droop effort; (c) states on dq -frame of the DG_1 unit (dq -coordinate of the filter voltage and current, respectively).

Figs. 6(a) and 7(a) depict the time marks for the events that define the stages. The variation of the parameter, primarily the nominal frequency variation, is illustrated in Fig. 7(b). Note that the states of DG_2 are not displayed in Figs. 6 and 7 because their performance is similar to that of the DG_1 unit.

Fig. 6(b) illustrates erratic behavior, which deteriorates when DG_2 is disconnected. This indicates that when the controller is designed solely based on $\mathcal{H}_\infty$ performance, the time response does not align with the desired behavior of the electrical microgrid. Conversely, when the $S(\alpha, d, r)$ constraints are integrated into the control synthesis technique, as demonstrated in Fig. 7, we observe a reference trajectory that aligns with the desired behavior of the electrical microgrid.

6.1.1. Comparison of the closed-loop system dynamics considering and disregarding D -stability constraints

When the GS full-order DOF controller is synthesized considering only (6), the eigenvalues of the closed-loop system's dynamic matrix fall outside the S region. This is demonstrated in Fig. 8 for various choices of ξ in Eq. (6). Consequently, this leads to an unsuitable time response for the application, as observed in Fig. 6(b).

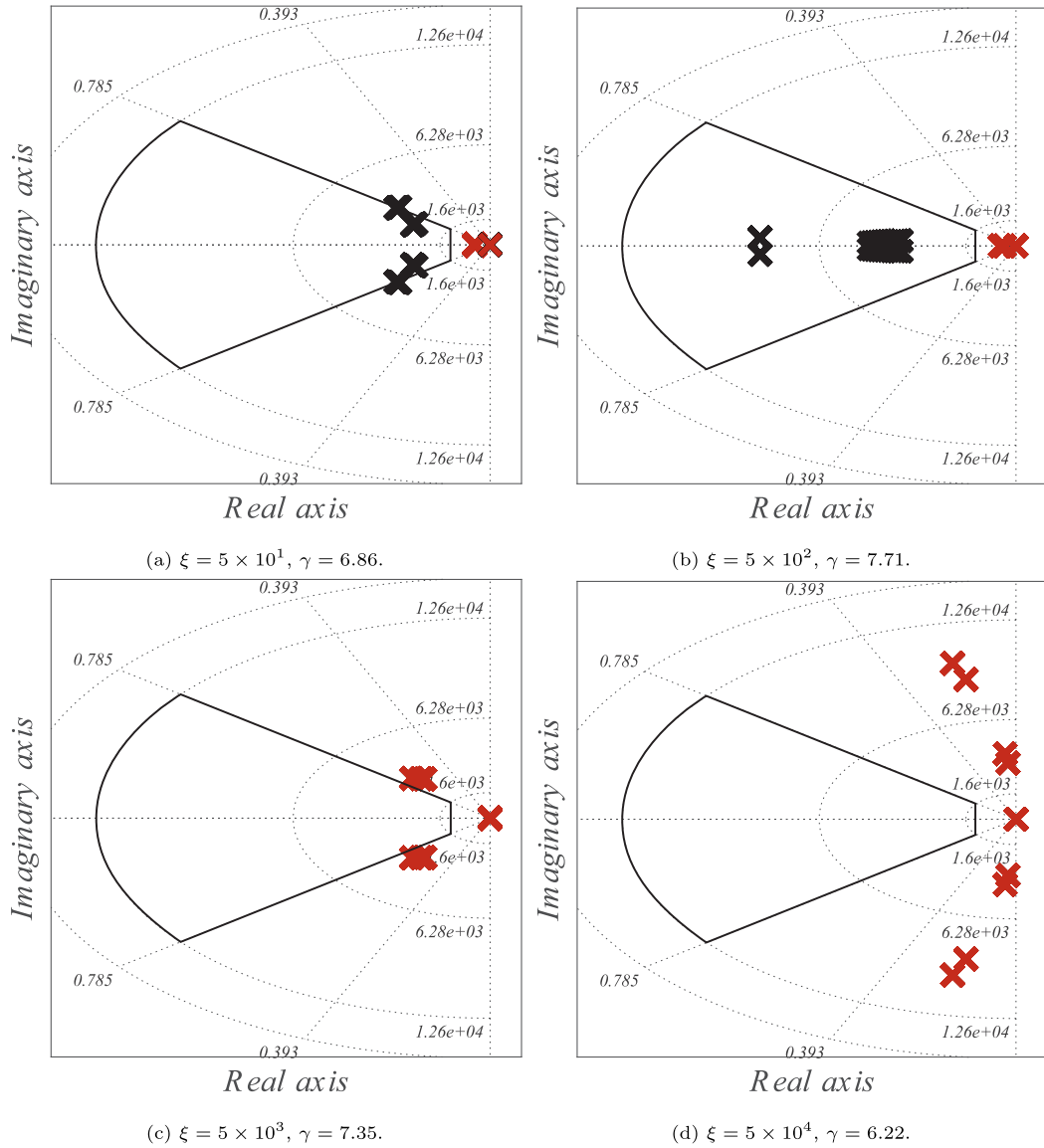


Fig. 8. Eigenvalues of the closed-loop system for frozen values of $\theta(t)$ obtained by taking 10 equidistant values between $\underline{\theta}$ to $\bar{\theta}$, and assuming distinct DOF controllers obtained by using four different values of the scalar parameter ξ on (6). The region bounded by a solid black line on the s -plane corresponds to the desired $S(a, d, r)$ region. The eigenvalues marked on red (■) represent an extrapolation of the limits established by the S region.

In contrast, when synthesizing the GS full-order DOF controller satisfying all the conditions in Theorem 1, the eigenvalues associated with the closed-loop system (for frozen values of $\theta(t)$) adhere with the constraints imposed by the S region. This compliance is evident in Fig. 9, resulting in a favorable time response, as depicted in Fig. 7.

6.2. Case study 2: Performance $J(T, f_\eta)$ based on sampling time sweep

In the simulations presented in this section, we consider the following performance index: $J = \int_{-\infty}^{\tau} x_e' x_e d\tau$. This index allows us to assess the quadratic cost of the tracking error, where a lower cost means better performance. The controller used corresponds to the same controller shown in Section 5.1. A sinusoidal wave with a frequency of 10^8 Hz was used as the disturbance signal $\bar{w}(t)$. The states $\bar{x}(t)$ exhibit frequency-dependent noise such that

$$\eta(f_\eta) = [10 \quad -10 \quad -1 \quad 1 \quad 0.1 \quad -0.1]' \sin(2\pi f_\eta).$$

The control input is considered piecewise constant in continuous-time, such that at each sample kT (s), $k = 1, 2, \dots$ of the output measurement $y(t)$, the microcontroller updates the control signal $u(t)$. The authors sweep the sampling time T and f_η . The index

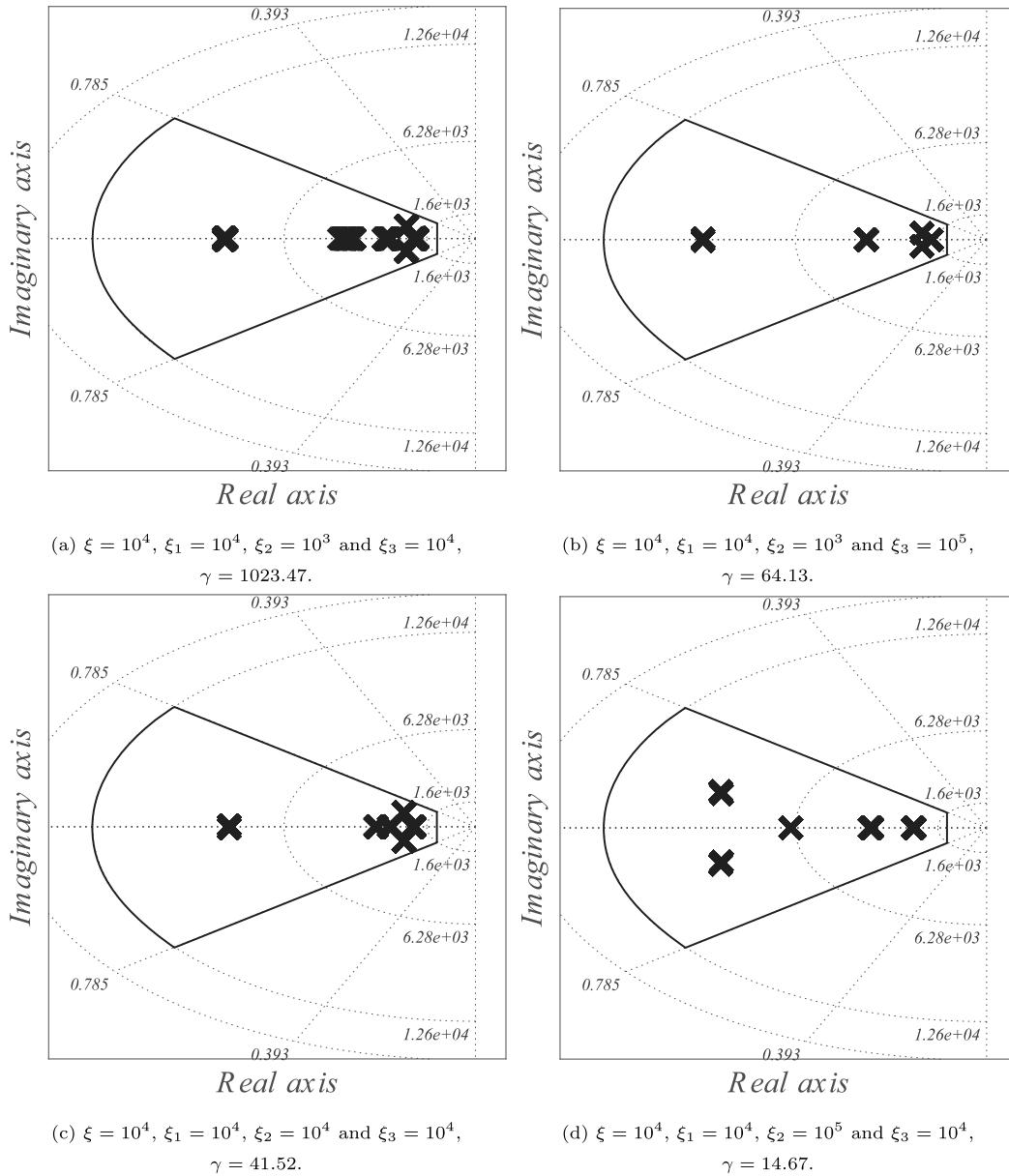
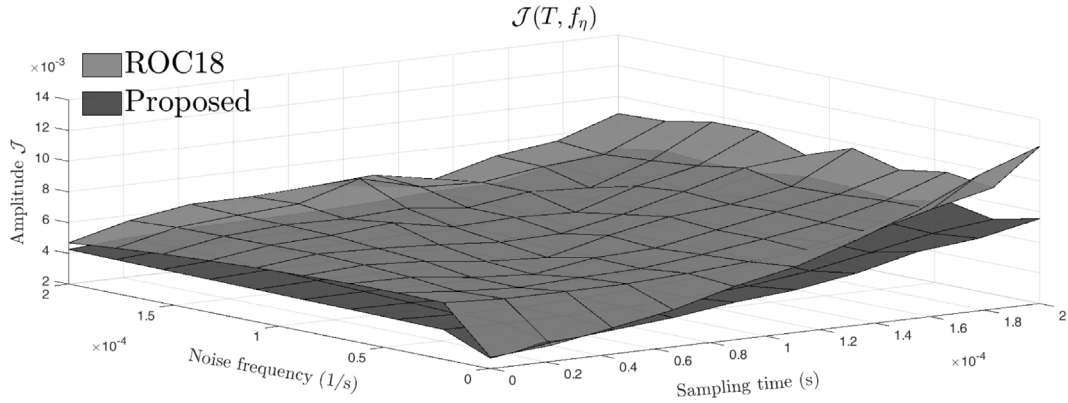


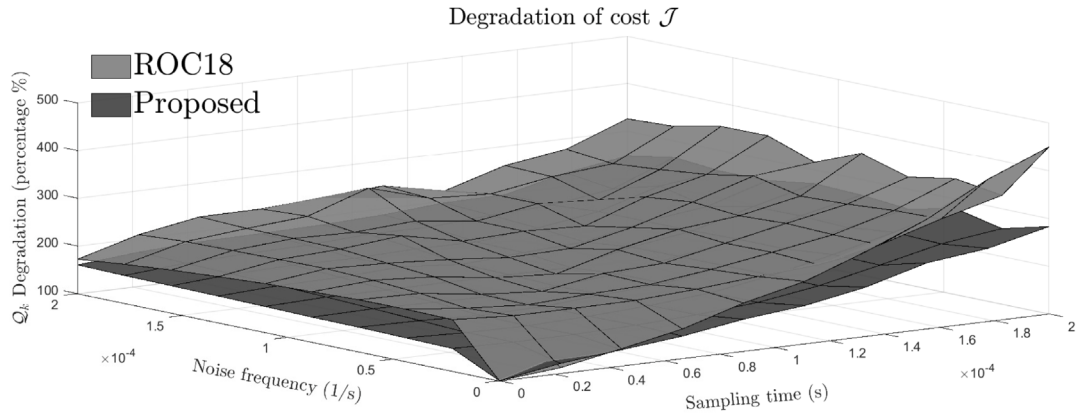
Fig. 9. Eigenvalues of the closed-loop system for frozen values of $\theta(t)$ obtained by taking 10 equidistant values between $\underline{\theta}$ to $\bar{\theta}$, and assuming distinct DOF controllers obtained by solving all conditions of Theorem 1 using $\xi = 10^4$ in (6) and 4 different combinations of the scalar parameters $[\xi_1, \xi_2, \xi_3]$. The region bounded by the solid black line on the s -plane corresponds to the desired $S(\alpha, d, r)$ region. The eigenvalues lie inside the S region. The scalar parameters were optimized to minimize the H_∞ guaranteed cost.

cost $J(T_i)$ was reported by taking 10 linear samples in the interval $T_i \in [10^{-7}, 10^{-4}]$ s and 10 logarithmic spaced samples in the interval $f_{\eta,i} \in [10^{-7}, 10^{-3}]$ s. The surface generated by computing the index J for the variation of both f_η and T is denoted by Q_k and is shown in Fig. 10. Each point of the surface was obtained to run the same temporal simulation that in Fig. 7, with the same events affecting the microgrid.

The result obtained by the proposed method was compared with the technique provided in [22], which is based solely on the H_∞ performance index. For all the studied sampling periods, the proposed controller demonstrates improved J performance. Fig. 10(b) illustrates the percentage increase relative to the ideal continuous-time noiseless case. It also demonstrates that the proposed controller maintains a consistent degradation percentage of performance compared with the controller in [22] across all ranges. For a more detailed comparison, a harmonic analysis, constructed using Fourier decomposition for fundamental and



(a) Performance $\mathcal{J}$ as a function of the sampling time. A logarithmic axis was used for the x -axis. Both the z and y -axis are linear.



(b) Percentage Degradation of Performance $\mathcal{J}$ relative to a noiseless continuous-time test. Linear scales were used for the x , y , and z -axis.

Fig. 10. Cost Q_k as a function of the sampling time. In Fig. 10(a), the cost Q_k is individually drawn for both [22] (ROC18) and the proposed controllers. In Fig. 10(b) the degradation percentage is plotted. The $\mathcal{J}(0.1 \mu\text{s})$ is considered as the nominal value.

harmonic frequencies,³ is presented in Fig. 11. This figure demonstrates that the proposed controller exhibits significantly lower amplitudes for the 2nd, 3rd, 4th, 5th, 7th, and 9th harmonics than the design technique presented in [22]. Consequently, by employing the proposed approach, a reduction of 11% in the total harmonic distortion can be achieved. In other words, the proposed controller is less susceptible to the adverse effects of noise and discretization errors.

As a final note on this experiment, it is important to highlight the success of the proposed approach in handling challenging phenomena such as additive noise in state signals (as there is inherent noise in semiconductor switching) and digital implementation considering the discretization error. These issues were effectively addressed by employing a suitable LPV model of the nonlinear system, accounting for exogenous disturbances and time-varying parameters. Subsequently, the closed-loop performance of the model was ensured by devising a gain-scheduled DOF controller using H_∞ norm and D -stability as performance criteria. It is worth noting that the DOF controller structure consistently yields superior results compared to static output feedback controllers, which proves advantageous in power electronics systems due to the influence of the aforementioned phenomena.

7. Conclusion

This paper introduced a new synthesis condition for a GS full-order DOF controller that achieves both H_∞ and D -stability for continuous-time LPV systems. These conditions are formulated as LMIs, and scalar searches can be employed to improve performance. The results were applied to a distributed power generation system based on micro-inverters, taking into account realistic operational characteristics. The results obtained demonstrated good transient performance, underscoring the importance of combining H_∞ performance and pole placement in terms of D -stability for such applications. Furthermore, comparisons with other strategies from the literature showcased the superiority of the proposed method. The next phase of this research involves applying the designed controller to a physical plant.

³ Find more information about harmonic analysis in [43].

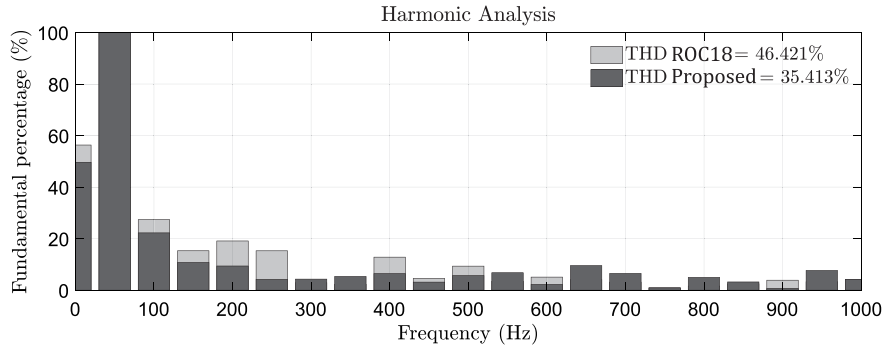


Fig. 11. Harmonic spectrum of the time-response of the closed-loop system. The amplitude is expressed as a percentage relative to the fundamental frequency, which is 50 Hz.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Matrices of the controller designed considering only the H_∞ performance

Fig. 8 presents a temporal simulation of the closed-loop system with a controller (obtained with $\xi = 5 \times 10^3$) with the following matrices:

$$\begin{aligned}
 A_{C0} &= \begin{bmatrix} -4650.0 & -253.0 & 1.81 \times 10^4 & -51.9 & -64.2 & -6.25 \\ 253.0 & -4650.0 & 51.9 & 1.81 \times 10^4 & 6.25 & -64.2 \\ -1030.0 & -128.0 & -3750.0 & 2050.0 & 60.9 & 0.613 \\ 128.0 & -1030.0 & -2050.0 & -3750.0 & -0.612 & 60.9 \\ -0.521 & -0.0338 & 0.0839 & -0.0138 & -0.183 & -8.3 \times 10^{-4} \\ 0.0338 & -0.521 & 0.0138 & 0.0839 & 8.3 \times 10^{-4} & -0.183 \end{bmatrix} \\
 A_{C1} &= \begin{bmatrix} 4.54 \times 10^{-10} & 3.73 \times 10^{-10} & 3.35 \times 10^{-9} & -6.5 \times 10^{-10} & -3.2 \times 10^{-12} & 3.08 \times 10^{-12} \\ 1.05 \times 10^{-9} & -2.15 \times 10^{-10} & -3.43 \times 10^{-9} & -3.76 \times 10^{-9} & -3.93 \times 10^{-12} & -2.84 \times 10^{-13} \\ -9.04 \times 10^{-6} & -1.23 \times 10^{-7} & 1.27 \times 10^{-5} & -9.38 \times 10^{-7} & -6.92 \times 10^{-9} & -3.27 \times 10^{-9} \\ -4.33 \times 10^{-7} & 1.11 \times 10^{-5} & 7.68 \times 10^{-7} & -1.84 \times 10^{-5} & 3.55 \times 10^{-9} & 1.81 \times 10^{-8} \\ 3.53 \times 10^{-13} & -7.59 \times 10^{-13} & -6.72 \times 10^{-13} & 6.42 \times 10^{-13} & -2.03 \times 10^{-16} & -6.44 \times 10^{-16} \\ -5.04 \times 10^{-13} & -2.53 \times 10^{-13} & 7.97 \times 10^{-13} & 2.27 \times 10^{-13} & -7.96 \times 10^{-16} & 7.29 \times 10^{-17} \end{bmatrix} \\
 B_{C0} &= \begin{bmatrix} 4650.0 & 551.0 & 1950.0 & 52.0 & 64.2 & 6.25 \\ -551.0 & 4650.0 & -52.0 & 1950.0 & -6.25 & 64.2 \\ 490.0 & 48.3 & -591.0 & -2340.0 & 3.88 & 1.58 \\ -48.3 & 490.0 & 2340.0 & -591.0 & -1.58 & 3.88 \\ -0.479 & 0.0338 & -0.0838 & 0.0141 & 0.183 & 8.26 \times 10^{-4} \\ -0.0338 & -0.479 & -0.0141 & -0.0838 & -8.26 \times 10^{-4} & 0.183 \end{bmatrix} \\
 B_{C1} &= \begin{bmatrix} -4.54 \times 10^{-10} & -3.73 \times 10^{-10} & -3.32 \times 10^{-9} & 6.55 \times 10^{-10} & 3.14 \times 10^{-12} & -3.07 \times 10^{-12} \\ -1.05 \times 10^{-9} & 2.15 \times 10^{-10} & 3.44 \times 10^{-9} & 3.73 \times 10^{-9} & 3.86 \times 10^{-12} & 6.66 \times 10^{-15} \\ 9.04 \times 10^{-6} & 1.22 \times 10^{-7} & -1.27 \times 10^{-5} & 9.23 \times 10^{-7} & 7.25 \times 10^{-9} & 3.51 \times 10^{-9} \\ 4.33 \times 10^{-7} & -1.11 \times 10^{-5} & -7.76 \times 10^{-7} & 1.84 \times 10^{-5} & -3.55 \times 10^{-9} & -1.78 \times 10^{-8} \\ -3.53 \times 10^{-13} & 7.59 \times 10^{-13} & 6.67 \times 10^{-13} & -6.4 \times 10^{-13} & 1.48 \times 10^{-16} & 5.86 \times 10^{-16} \\ 5.04 \times 10^{-13} & 2.53 \times 10^{-13} & -7.99 \times 10^{-13} & -2.19 \times 10^{-13} & 8.03 \times 10^{-16} & -8.93 \times 10^{-17} \end{bmatrix} \\
 C_{C0} &= \begin{bmatrix} -0.527 & -0.059 & -4.95 & 2.42 & 0.088 & 0.00298 \\ 0.059 & -0.527 & -2.42 & -4.95 & -0.00298 & 0.088 \end{bmatrix} \\
 C_{C1} &= \begin{bmatrix} -1.22 \times 10^{-8} & -1.67 \times 10^{-10} & 1.71 \times 10^{-8} & -1.27 \times 10^{-9} & -9.34 \times 10^{-12} & -4.42 \times 10^{-12} \\ -5.85 \times 10^{-10} & 1.49 \times 10^{-8} & 1.04 \times 10^{-9} & -2.49 \times 10^{-8} & 4.8 \times 10^{-12} & 2.44 \times 10^{-11} \end{bmatrix}
 \end{aligned}$$

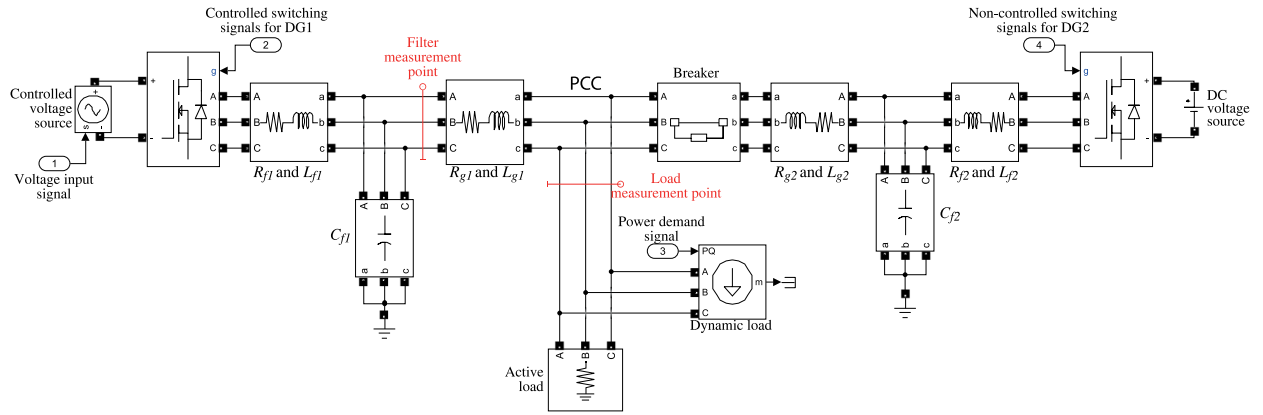


Fig. B.1. Implementation environment for simulated experiments.

$$D_{C0} = \begin{bmatrix} 0.801 & -0.0491 & -0.812 & -3.2 & -4.68 \times 10^{-4} & -1.29 \times 10^{-5} \\ 0.0491 & 0.801 & 3.2 & -0.812 & 1.28 \times 10^{-5} & -4.68 \times 10^{-4} \end{bmatrix}$$

$$D_{C1} = \begin{bmatrix} 1.22 \times 10^{-8} & 1.65 \times 10^{-10} & -1.71 \times 10^{-8} & 1.25 \times 10^{-9} & 9.79 \times 10^{-12} & 4.74 \times 10^{-12} \\ 5.85 \times 10^{-10} & -1.49 \times 10^{-8} & -1.05 \times 10^{-9} & 2.49 \times 10^{-8} & -4.8 \times 10^{-12} & -2.41 \times 10^{-11} \end{bmatrix}$$

Appendix B. Simulation setup based on MATLAB electronic specialized software

To better simulate electrical systems, specialized software was utilized. The *Simscape Specialized Power Systems toolbox* from the MATLAB environment was used to implement the simulation setup. The scheme implemented in *Simulink* is shown in Fig. B.1, indicating the correspondence between its components and those of Fig. 3.

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